

Kafka Integration with C#

Task 1 — Chat App using Kafka

Step 1 — Install NuGet Package

Create a C# Console Application and install Confluent.Kafka:

```
Install-Package Confluent.Kafka
```

Step 2 — Producer Code

Sends messages to the 'chat-message' Kafka topic.

```
using System; using
Confluent.Kafka;

class Producer
{
    static void Main(string[] args)
    {
        var config = new ProducerConfig { BootstrapServers = "localhost:9092" };

        using (var producer = new ProducerBuilder<Null, string>(config).Build())
        {
            Console.WriteLine("Type message and press Enter. Ctrl+C to
            exit.");
            while (true)
            {
                string message = Console.ReadLine();
                producer.Produce("chat-message",
                    new Message<Null, string> { Value = message },
                    (dr) => {
                        if (dr.Error.Code != ErrorCode.NoError)
                            Console.WriteLine($"Error: {dr.Error.Reason}");
                        else
                            Console.WriteLine($"Sent to: {dr.TopicPartitionOffset}");
                        producer.Flush(TimeSpan.FromSeconds(1));
                    });
            }
        }
    }
}
```

Step 3 — Consumer Code

Consumes messages from the 'chat-message' topic in the command prompt.

```
using System; using
Confluent.Kafka;

class Consumer
{
    static void Main(string[] args)
    {
        var config = new ConsumerConfig
        {
            GroupId = "chat-group",
            BootstrapServers = "localhost:9092",
            AutoOffsetReset = AutoOffsetReset.Earliest
        };

        using (var consumer = new ConsumerBuilder<Ignore, string>(config).Build())
        {
            consumer.Subscribe("chat-message");
            Console.WriteLine("Listening for messages...");
            while (true)
            {
                var cr = consumer.Consume();
                Console.WriteLine($"Received: {cr.Message.Value}");
            }
        }
    }
}
```

```

    }
}
}
}
}

```

Step 4 — Start Zookeeper `zookeeper-server-start.bat`

`..\..\config\zookeeper.properties` [Screenshot](#) — Zookeeper

Running:

```

C:\Windows\System32\cmd.exe - zookeeper-server-start.bat ..\..\config\zookeeper.properties
direct buffers. (org.apache.zookeeper.server.NIOServerCnxnFactory)
[2020-12-23 13:17:15,770] INFO binding to port 0.0.0.0/0.0.0.0:2181 (org.apache.zookeeper.server.NIOServerCnxnFactory)
[2020-12-23 13:17:15,807] INFO zookeeper.snapshotSizeFactor = 0.33 (org.apache.zookeeper.server.ZKDatabase)
[2020-12-23 13:17:15,815] INFO Snapshotting: 0x0 to H:\Kafkakafka_2.12-2.7.0zookeeper\version-2\snapshot.0 (org.apache.zookeeper.server.persistence.FileTxnSnapshotLog)
[2020-12-23 13:17:15,842] INFO Snapshotting: 0x0 to H:\Kafkakafka_2.12-2.7.0zookeeper\version-2\snapshot.0 (org.apache.zookeeper.server.persistence.FileTxnSnapshotLog)
[2020-12-23 13:17:15,896] INFO Using checkIntervalMs=60000 maxPerMinute=10000 (org.apache.zookeeper.server.ContainerManager)
[2020-12-23 13:18:45,133] INFO Creating new log file: log.1 (org.apache.zookeeper.server.persistence.FileTxnLog)
[2020-12-23 13:21:08,152] WARN Unable to read additional data from client sessionid 0x1001e9b51ca0001, likely client has closed socket (org.apache.zookeeper.server.NIOServerCnxn)

```

Step 5 — Start Kafka Server `kafka-server-start.bat`

`..\..\config\server.properties` [Screenshot](#) — Kafka

Server Running:

```

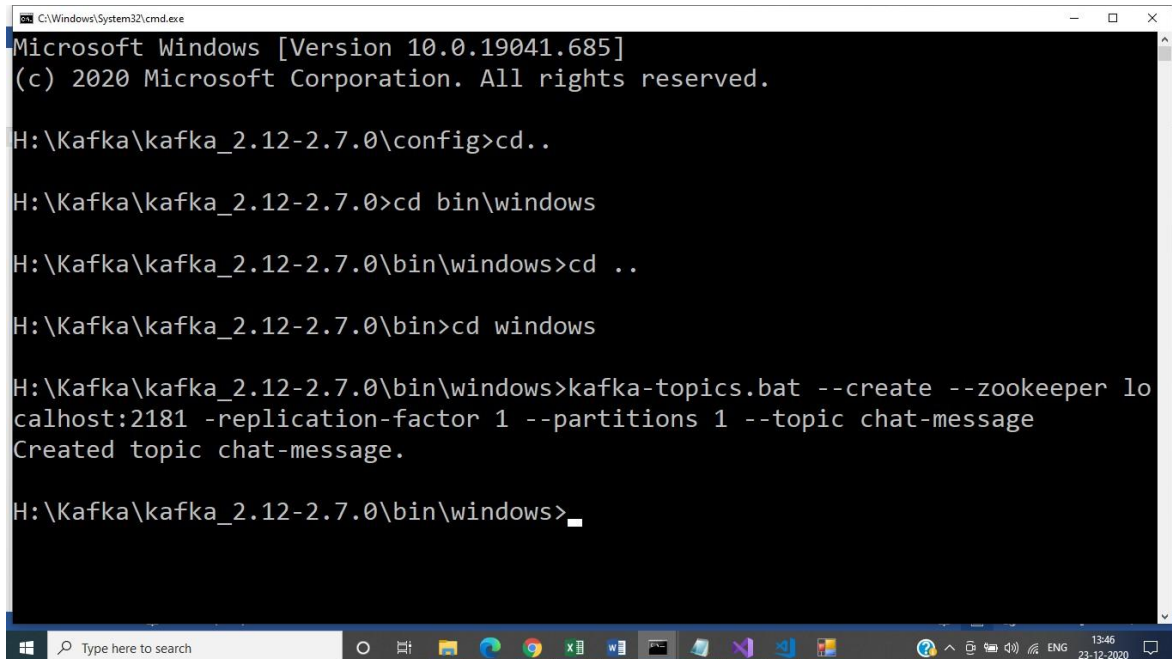
Command Prompt - kafka-server-start.bat ..\..\config\server.properties
[2020-12-23 13:24:15,159] INFO [GroupMetadataManager brokerId=0] Finished loading offsets and group metadata from __consumer_offsets-18 in 26 milliseconds, of which 26 milliseconds was spent in the scheduler. (kafka.coordinator.group.GroupMetadataManager)
[2020-12-23 13:24:15,160] INFO [GroupMetadataManager brokerId=0] Finished loading offsets and group metadata from __consumer_offsets-21 in 26 milliseconds, of which 25 milliseconds was spent in the scheduler. (kafka.coordinator.group.GroupMetadataManager)
[2020-12-23 13:24:15,160] INFO [GroupMetadataManager brokerId=0] Finished loading offsets and group metadata from __consumer_offsets-24 in 26 milliseconds, of which 26 milliseconds was spent in the scheduler. (kafka.coordinator.group.GroupMetadataManager)
[2020-12-23 13:24:15,161] INFO [GroupMetadataManager brokerId=0] Finished loading offsets and group metadata from __consumer_offsets-27 in 26 milliseconds, of which 26 milliseconds was spent in the scheduler. (kafka.coordinator.group.GroupMetadataManager)
[2020-12-23 13:24:15,161] INFO [GroupMetadataManager brokerId=0] Finished loading offsets and group metadata from __consumer_offsets-30 in 26 milliseconds, of which 26 milliseconds was spent in the scheduler. (kafka.coordinator.group.GroupMetadataManager)
[2020-12-23 13:24:15,162] INFO [GroupMetadataManager brokerId=0] Finished loading offsets and group metadata from __consumer_offsets-33 in 26 milliseconds, of which 26 milliseconds was spent in the scheduler. (kafka.coordinator.group.GroupMetadataManager)
[2020-12-23 13:24:15,164] INFO [GroupMetadataManager brokerId=0] Finished loading offsets and group metadata from __consumer_offsets-36 in 27 milliseconds, of which 26 milliseconds was spent in the scheduler. (kafka.coordinator.group.GroupMetadataManager)
[2020-12-23 13:24:15,165] INFO [GroupMetadataManager brokerId=0] Finished loading offsets and group metadata from __consumer_offsets-39 in 28 milliseconds, of which 27 milliseconds was spent in the scheduler. (kafka.coordinator.group.GroupMetadataManager)
[2020-12-23 13:24:15,165] INFO [GroupMetadataManager brokerId=0] Finished loading offsets and group metadata from __consumer_offsets-42 in 27 milliseconds, of which 27 milliseconds was spent in the scheduler. (kafka.coordinator.group.GroupMetadataManager)
[2020-12-23 13:24:15,166] INFO [GroupMetadataManager brokerId=0] Finished loading offsets and group metadata from __consumer_offsets-45 in 27 milliseconds, of which 27 milliseconds was spent in the scheduler. (kafka.coordinator.group.GroupMetadataManager)
[2020-12-23 13:24:15,167] INFO [GroupMetadataManager brokerId=0] Finished loading offsets and group metadata from __consumer_offsets-48 in 27 milliseconds, of which 27 milliseconds was spent in the scheduler. (kafka.coordinator.group.GroupMetadataManager)
[2020-12-23 13:24:15,262] INFO [GroupCoordinator 0]: Preparing to rebalance group console-consumer-37318 in state PreparingRebalance with old generation 0 (__consumer_offsets-13) (reason: Adding new member consumer-console-consumer-37318-1-5b0fd435-1eeb-4e28-90b4-ba45b3fbc9cb with group instance id None) (kafka.coordinator.group.GroupCoordinator)
[2020-12-23 13:24:15,284] INFO [GroupCoordinator 0]: Stabilized group console-consumer-37318 generation 1 (__consumer_offsets-13) (kafka.coordinator.group.GroupCoordinator)
[2020-12-23 13:24:15,310] INFO [GroupCoordinator 0]: Assignment received from leader for group console-consumer-37318 for generation 1 (kafka.coordinator.group.GroupCoordinator)

```

Step 6 — Create Topic

```
kafka-topics.bat --create --zookeeper localhost:2181 --  
replication-factor 1 --partitions 1 --topic chat-message
```

Screenshot — Topic Created:

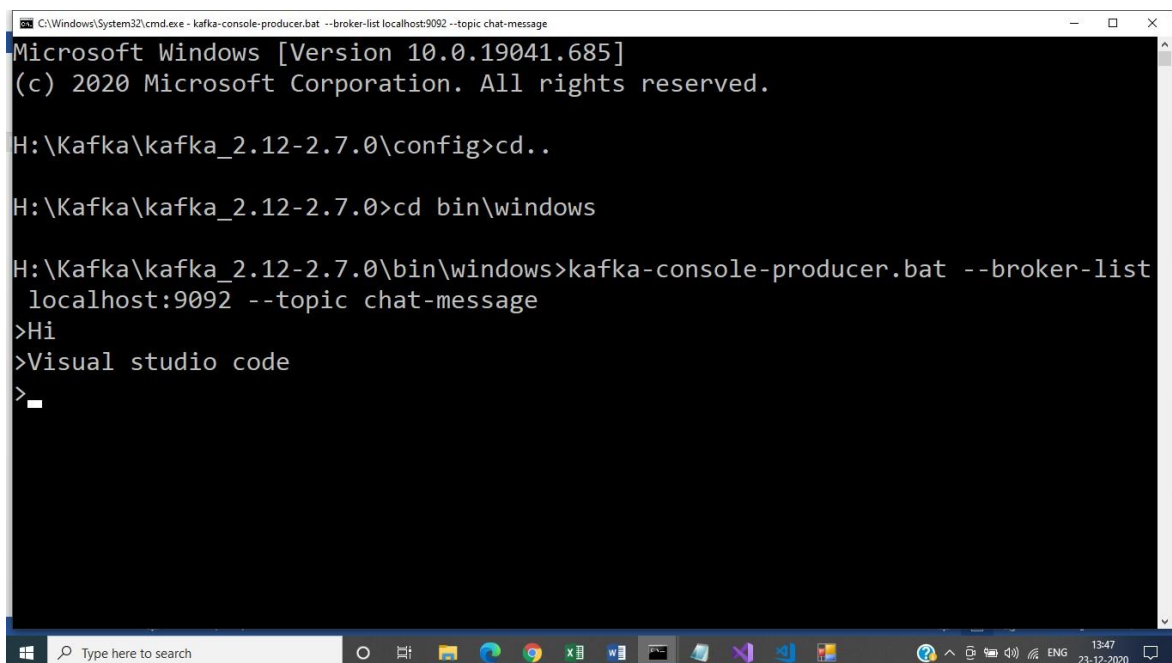


```
C:\Windows\System32\cmd.exe  
Microsoft Windows [Version 10.0.19041.685]  
(c) 2020 Microsoft Corporation. All rights reserved.  
  
H:\Kafka\kafka_2.12-2.7.0\config>cd..  
  
H:\Kafka\kafka_2.12-2.7.0>cd bin\windows  
  
H:\Kafka\kafka_2.12-2.7.0\bin\windows>cd ..  
  
H:\Kafka\kafka_2.12-2.7.0\bin>cd windows  
  
H:\Kafka\kafka_2.12-2.7.0\bin\windows>kafka-topics.bat --create --zookeeper lo  
calhost:2181 -replication-factor 1 --partitions 1 --topic chat-message  
Created topic chat-message.  
  
H:\Kafka\kafka_2.12-2.7.0\bin\windows>_
```

Step 7 — Produce Messages

```
kafka-console-producer.bat --broker-list localhost:9092 --topic chat-message
```

Screenshot — Producer sending messages:



```
C:\Windows\System32\cmd.exe - kafka-console-producer.bat --broker-list localhost:9092 --topic chat-message  
Microsoft Windows [Version 10.0.19041.685]  
(c) 2020 Microsoft Corporation. All rights reserved.  
  
H:\Kafka\kafka_2.12-2.7.0\config>cd..  
  
H:\Kafka\kafka_2.12-2.7.0>cd bin\windows  
  
H:\Kafka\kafka_2.12-2.7.0\bin\windows>kafka-console-producer.bat --broker-list  
localhost:9092 --topic chat-message  
>Hi  
>Visual studio code  
>_
```

Step 8 — Consume Messages in Command Prompt

```
kafka-console-consumer.bat --bootstrap-server localhost:9092  
--topic chat-message --from-beginning
```

Screenshot — Consumer receiving messages:

```
Command Prompt - kafka-console-consumer.bat --bootstrap-server localhost:9092 --topic chat-message --from-beginning
Microsoft Windows [Version 10.0.19041.685]
(c) 2020 Microsoft Corporation. All rights reserved.

C:\Users\Shrivalli>h:

H:\>cd kafka
H:\Kafka>cd kafka*
H:\Kafka\kafka_2.12-2.7.0>cd windows
The system cannot find the path specified.
H:\Kafka\kafka_2.12-2.7.0>cd bin
H:\Kafka\kafka_2.12-2.7.0\bin>cd windows
H:\Kafka\kafka_2.12-2.7.0\bin\windows>kafka-console-consumer.bat --bootstrap-server localhost:9092 --topic chat-message --from-beginning
welcome
welcome
This is the first message
welcome
This is the first message
HI
Visual studio code
```

Task 2 — Chat App using C# Windows Forms + Kafka

Create a chat application using a C# Windows Forms Application that uses Kafka to send messages and consume them in different client applications.

Step 1 — Create Windows Forms Project

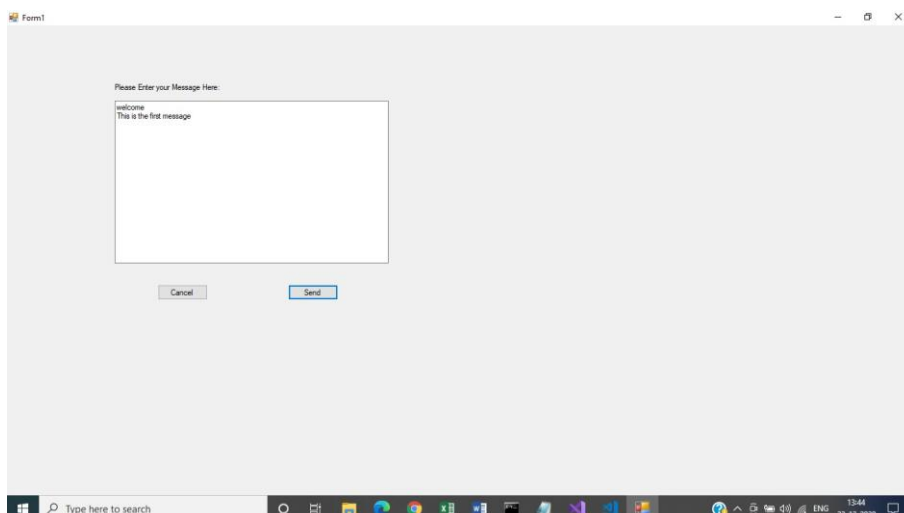
Create a new Windows Forms App (.NET Framework) in Visual Studio and install Confluent.Kafka:

```
Install-Package Confluent.Kafka
```

Step 2 — Form Design

```
Controls added to Form1:
Label    : Text = "Please Enter your Message Here:"
TextBox  : Name = txtMessage, Multiline = True
Button   : Name = btnSend, Text = "Send"
Button   : Name = btnCancel, Text = "Cancel"
```

Screenshot — Windows Form UI:



Step 3 — Form1.cs Code (Producer)

```
using System;
```

```

using System.Windows.Forms;
using Confluent.Kafka;

namespace KafkaChatApp
{
    public partial class Form1 : Form
    {
        private IProducer<Null, string> _producer;

        public Form1()
        {
            InitializeComponent();
            var config = new ProducerConfig { BootstrapServers = "localhost:9092" };
            _producer = new ProducerBuilder<Null, string>(config).Build();

            private async void btnSend_Click(object sender, EventArgs e)
            {
                string msg = txtMessage.Text.Trim();
                if (string.IsNullOrEmpty(msg)) return;

                await _producer.ProduceAsync("chat-message",
                    new Message<Null, string> { Value = msg });

                MessageBox.Show("Message sent!", "Success",
                    MessageBoxButtons.OK, MessageBoxIcon.Information);
                txtMessage.Clear();
            }

            private void btnCancel_Click(object sender, EventArgs e)
            {
                txtMessage.Clear();
            }

            protected override void OnFormClosing(FormClosingEventArgs e)
            {
                _producer?.Dispose();
                base.OnFormClosing(e);
            }
        }
    }
}

```

Step 4 — Consumer (separate app or Form)

The consumer can be the command-prompt consumer from Task 1, or a second Windows Forms app using a `BackgroundWorker` to poll messages:

```

// Form2 - Receiver (Consumer)
using System.ComponentModel;
using System.Windows.Forms;
using Confluent.Kafka;

public partial class Form2 : Form
{
    private BackgroundWorker _worker = new BackgroundWorker();

    public Form2()
    {
        InitializeComponent();
        _worker.DoWork += Worker_DoWork;
        _worker.RunWorkerAsync();
    }

    private void Worker_DoWork(object sender, DoWorkEventArgs e)
    {
        var config = new ConsumerConfig
        {
            GroupId = "winforms-group",
            BootstrapServers = "localhost:9092",
            AutoOffsetReset = AutoOffsetReset.Earliest
        };
    }
}

```

```

        using (var consumer = new ConsumerBuilder<Ignore, string>(config).Build())
        {
            consumer.Subscribe("chat-message");
            while (true)
            {
                var cr = consumer.Consume();
                Invoke(new Action(() =>
                    lstMessages.Items.Add(cr.Message.Value)));
            }
        }
    }
}

```

Output — Messages received from the Windows Form

Messages typed and sent from the Windows Form appear in the consumer. Screenshot below shows the consumer receiving all messages:

```

C:\Users\Shrivalli>h:
H:\>cd kafka
H:\Kafka>cd kafka*
H:\Kafka\kafka_2.12-2.7.0>cd windows
The system cannot find the path specified.
H:\Kafka\kafka_2.12-2.7.0>cd bin
H:\Kafka\kafka_2.12-2.7.0\bin>cd windows
H:\Kafka\kafka_2.12-2.7.0\bin\windows>kafka-console-consumer.bat --bootstrap-server localhost:9092 --topic chat-message --from-beginning
welcome
welcome
This is the first message
welcome
This is the first message
Hi
Visual studio code

```